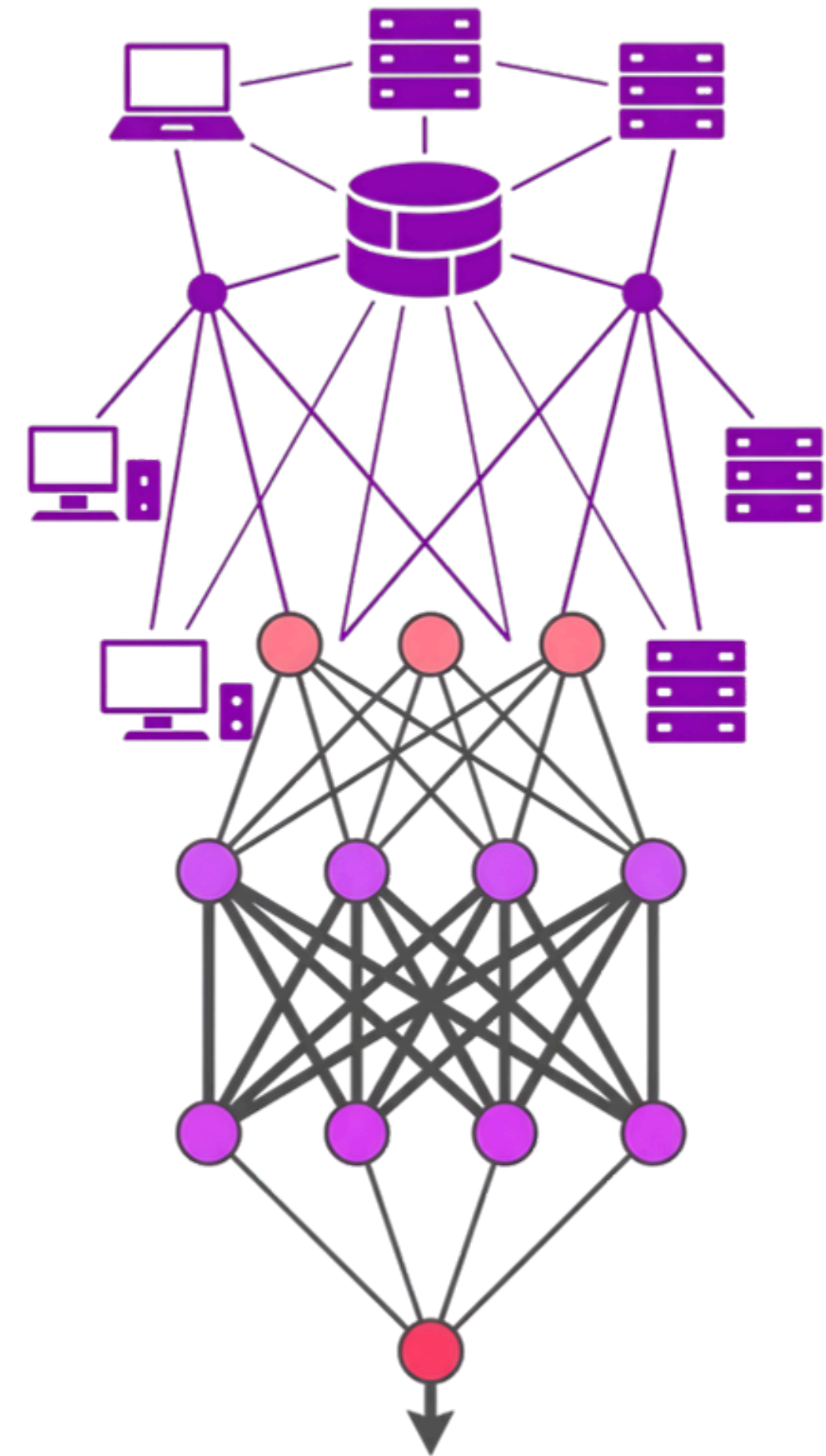


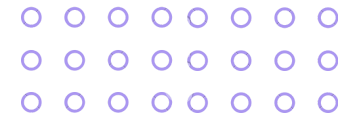
SYSTEMATIC SYNTHESIS AND BENCHMARKING OF DEEP LEARNING ARCHITECTURES FOR NETWORK INTRUSION DETECTION

Presented By :

Oussama BEN SASSI



PRESENTATION OUTLINE



01 LITERATURE OVERVIEW AND MAJOR DATASET DISTRIBUTION

02 ML AND DL PARADIGMS IN SOTA

03 BRIEF MACHINE LEARNING BASELINES REVIEW

04 EXHAUSTIVE REVIEW ON SUPERVISED AND UNSUPERVISED DEEP LEARNING ARCHITECTURES

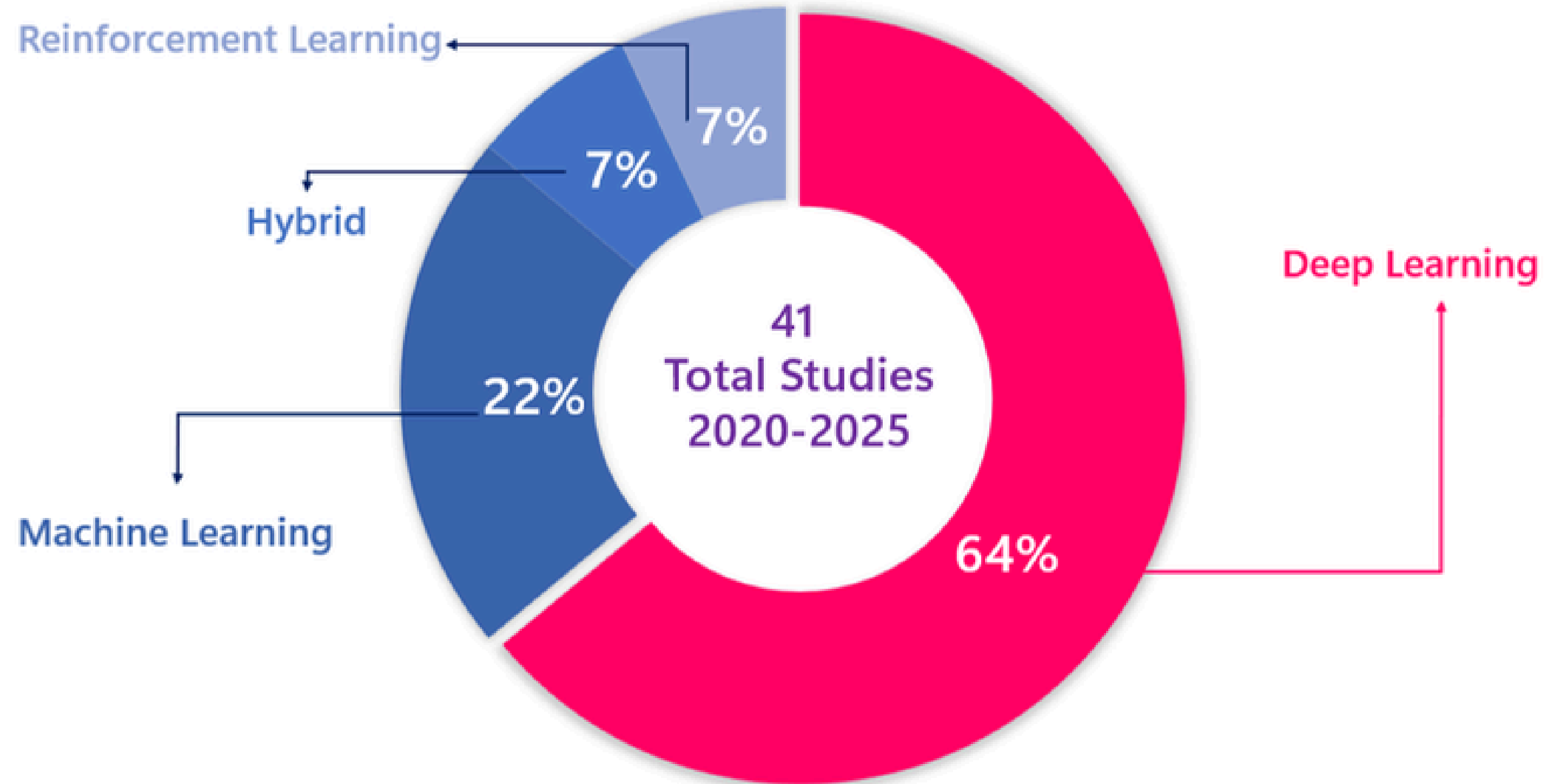
05 COMPARATIVE SUMMARY OF REVIEWED IDS ARCHITECTURES

06 IDENTIFIED RESEARCH GAPS

07 CONCLUSION AND THE PATH FORWARD FOR DL-IDS

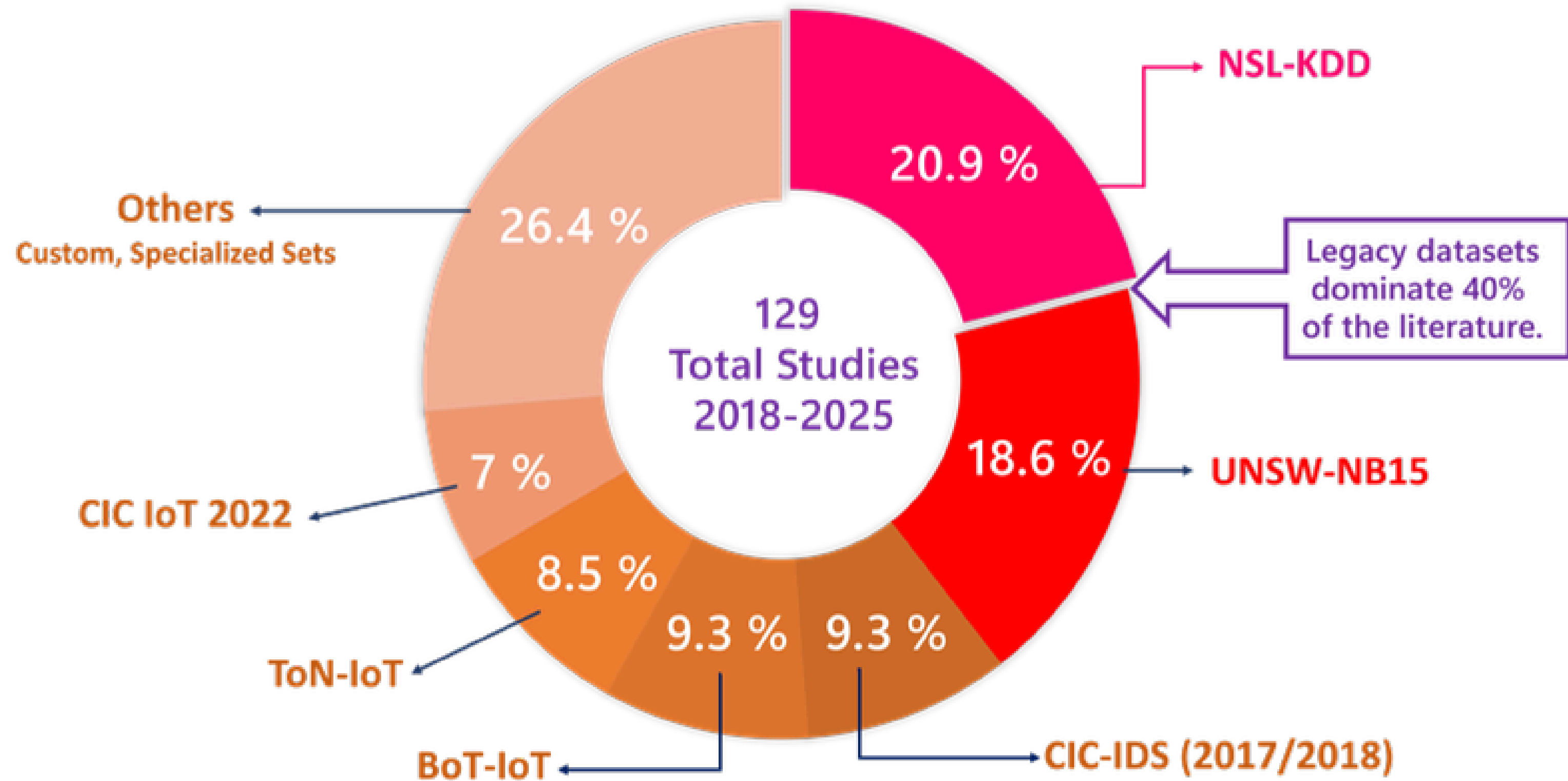


LITERATURE DISTRIBUTION



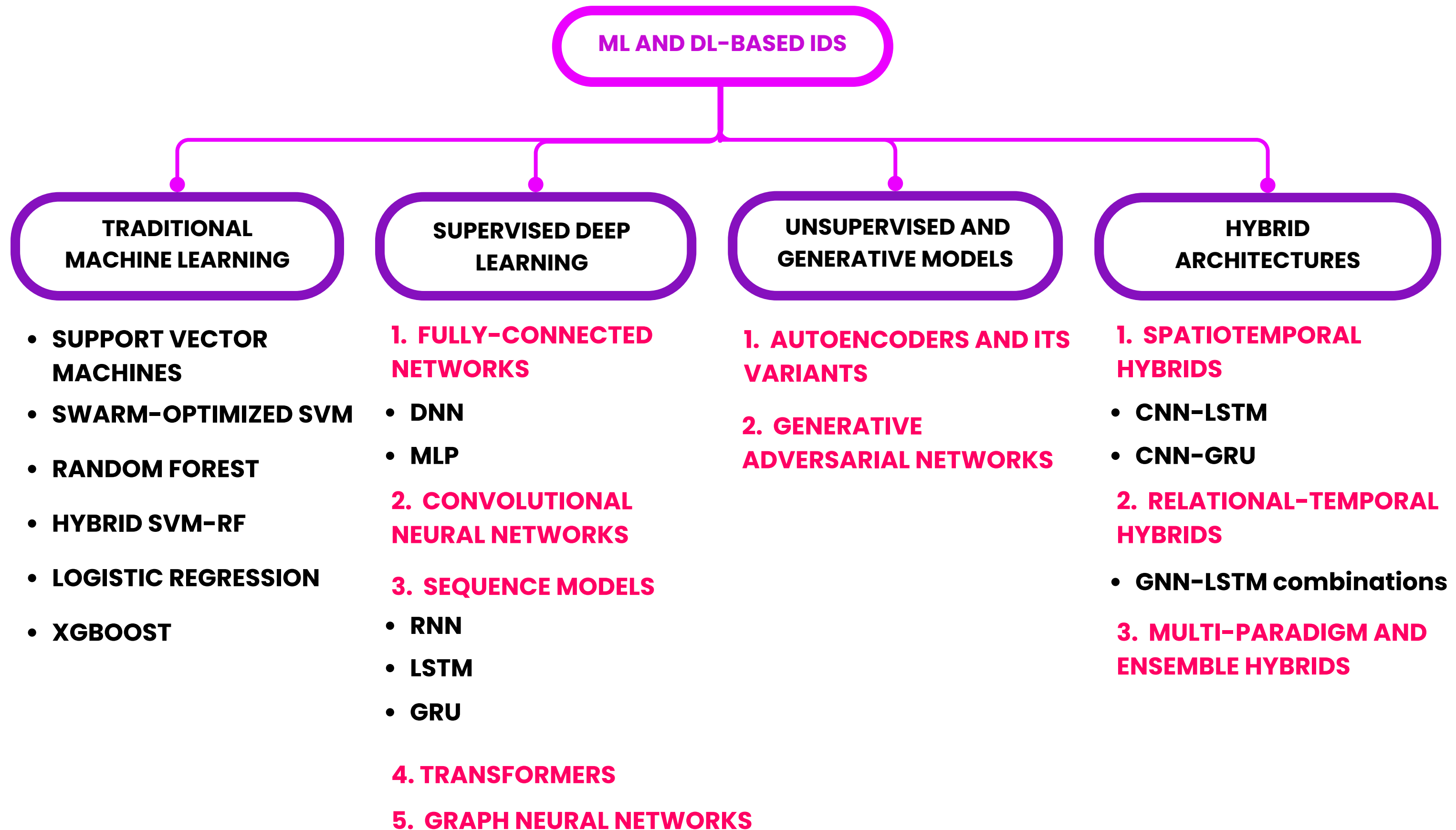
[ZMAIMITA ET AL., 2025]

MAJOR DATASET DISTRIBUTION

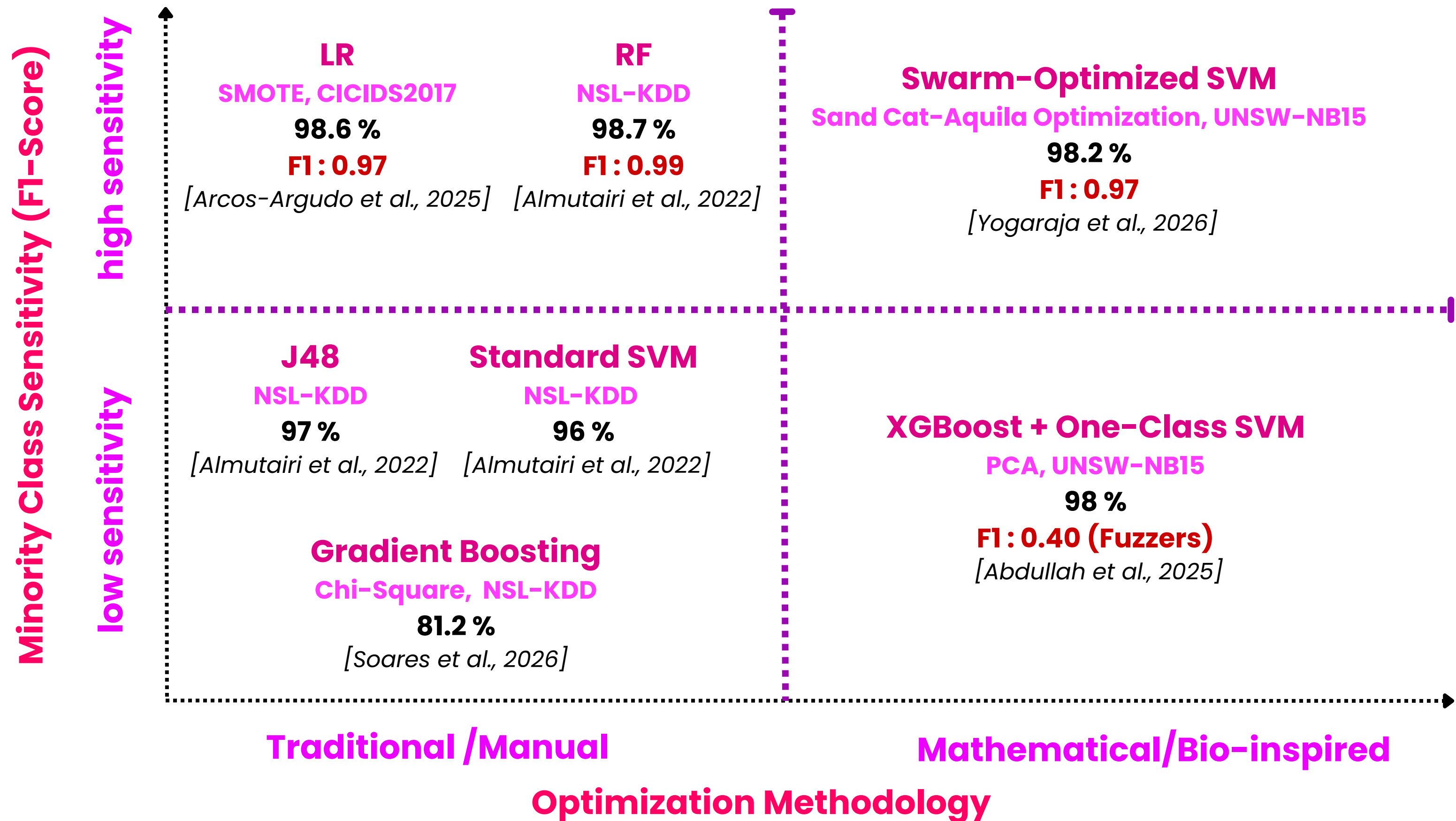


[OGUNSEYI ET AL., 2026]

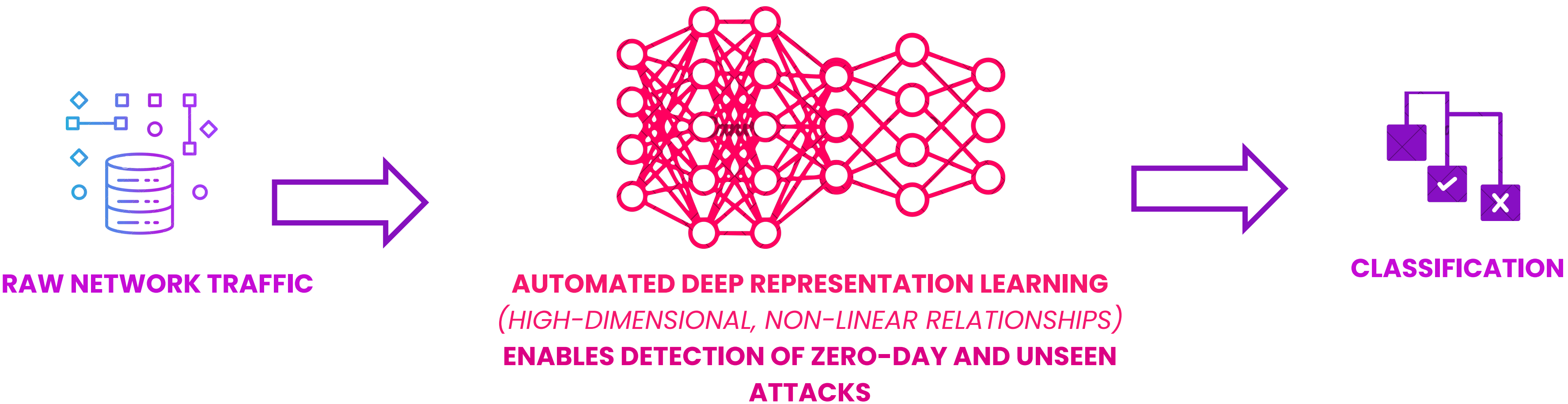
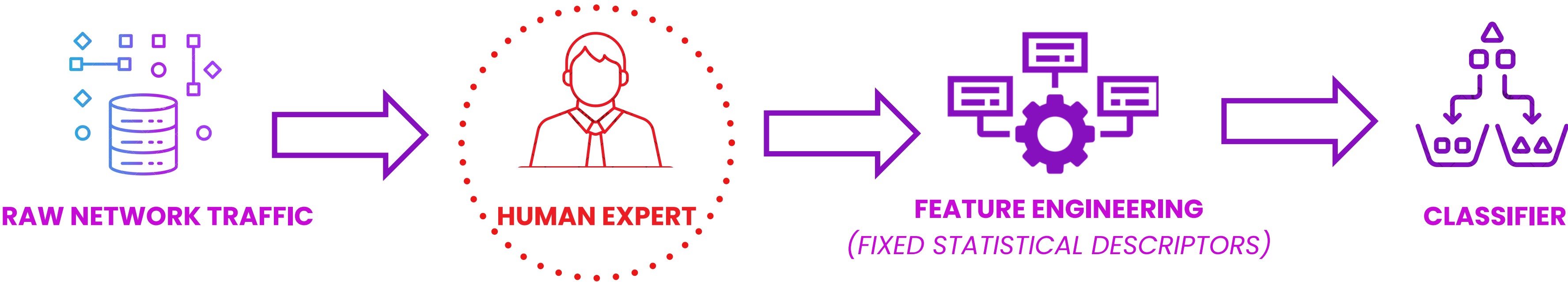
ML AND DL PARADIGMS IN SOTA



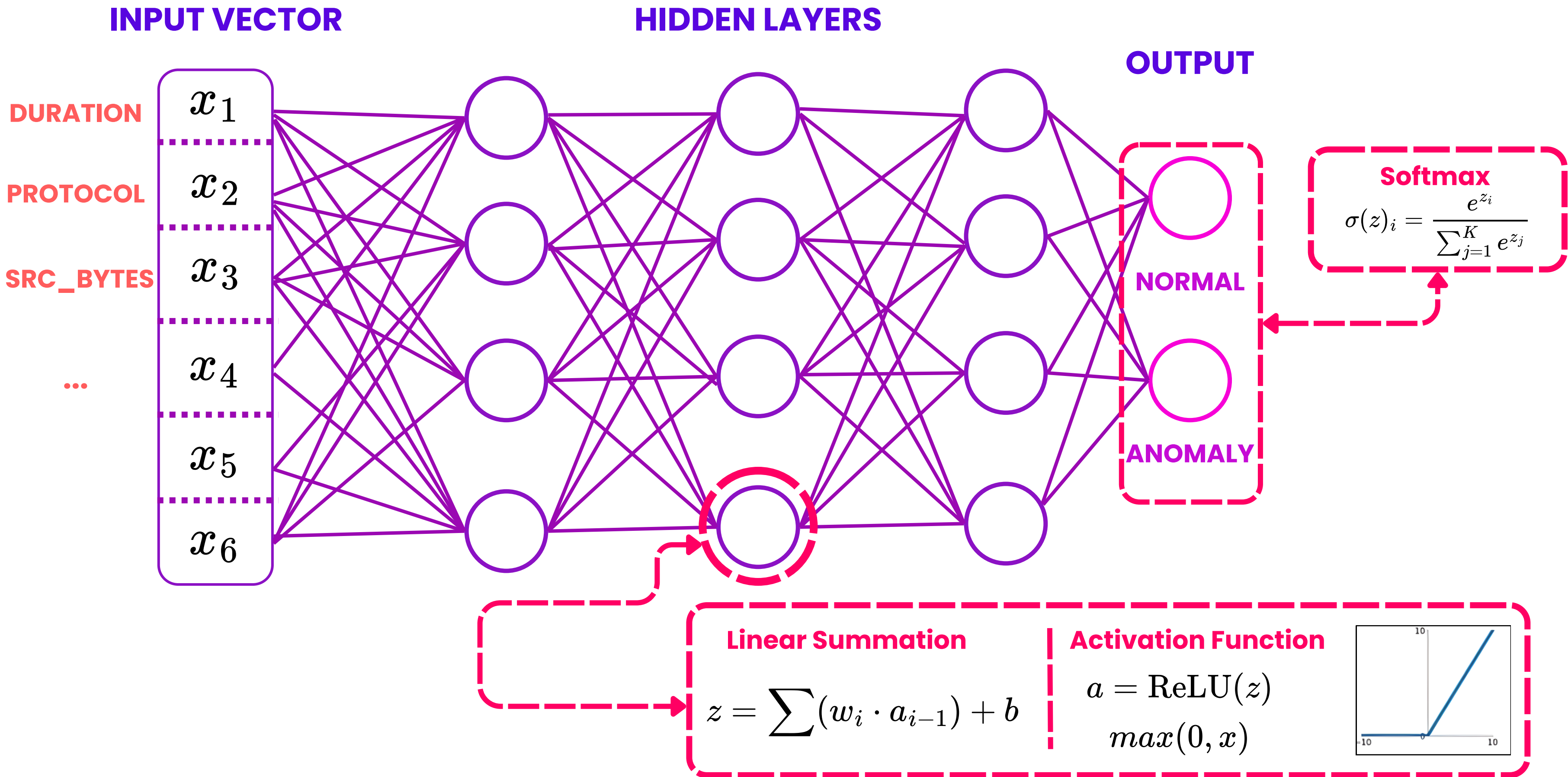
MACHINE LEARNING BASELINES



ML LIMITATION



DEEP NEURAL NETWORKS ARCHITECTURAL PARADIGM



FCN IN LITERATURE

Binary classification

DNN, NF-UNSW-NB15

98 %

[Shaker et al., 2023]

Internal Optimization

OSNN (GeLU/LeakyReLU, L2)

CSE-CIC-IDS-2018

99.53 %

stable convergence

[Alsubaei, 2025]

Multiclass classification

MLP, CIC-IDS-2017

97 %

unstable convergence

[Ali et al., 2025]

Latent Paradigm

AE+MLP, NSL-KDD

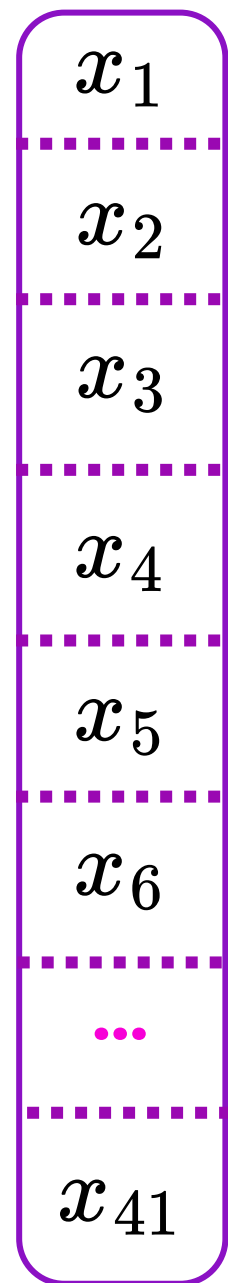
86 %

Stability through latent features

[Hussein & Répás, 2026]

CNN PARADIGM

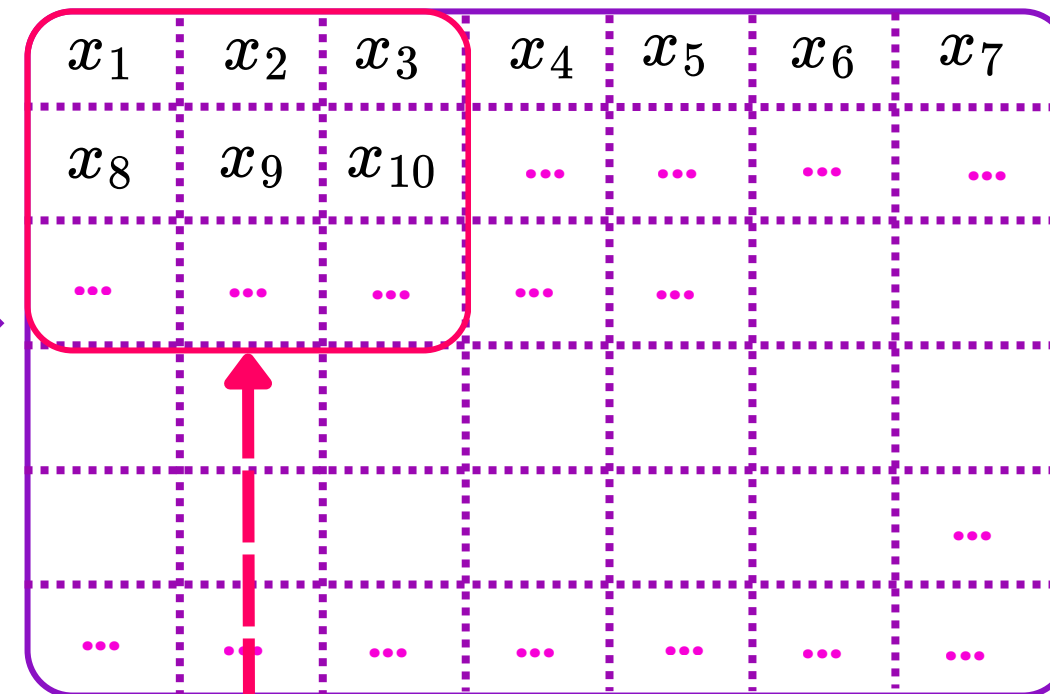
1D
INPUT VECTOR



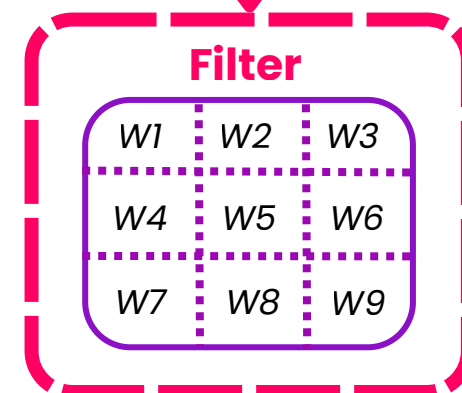
SPATIAL RESHAPING
 $\mathbb{R}^{41} \rightarrow \mathbb{R}^{6 \times 7 \times 1}$



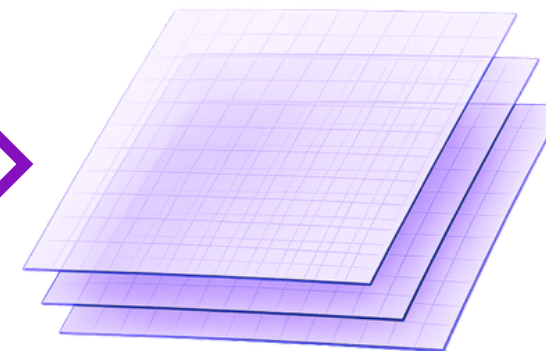
2D
MATRIX



$$z = \mathbf{W} * \mathbf{x} + b$$



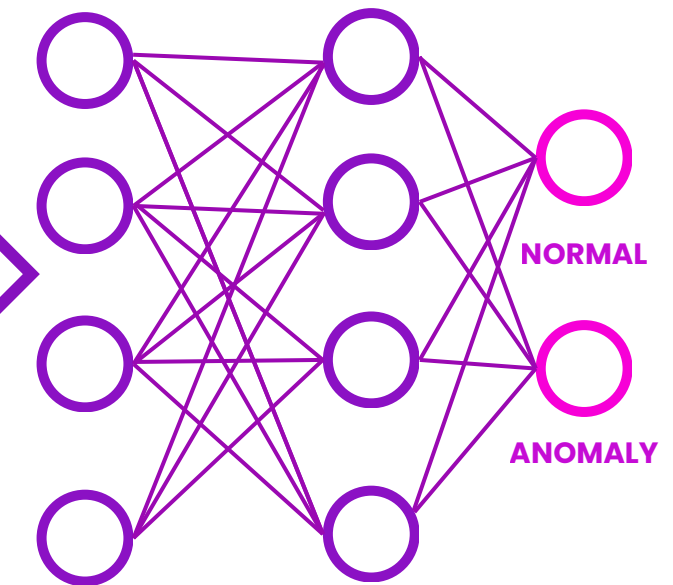
FEATURE MAPS



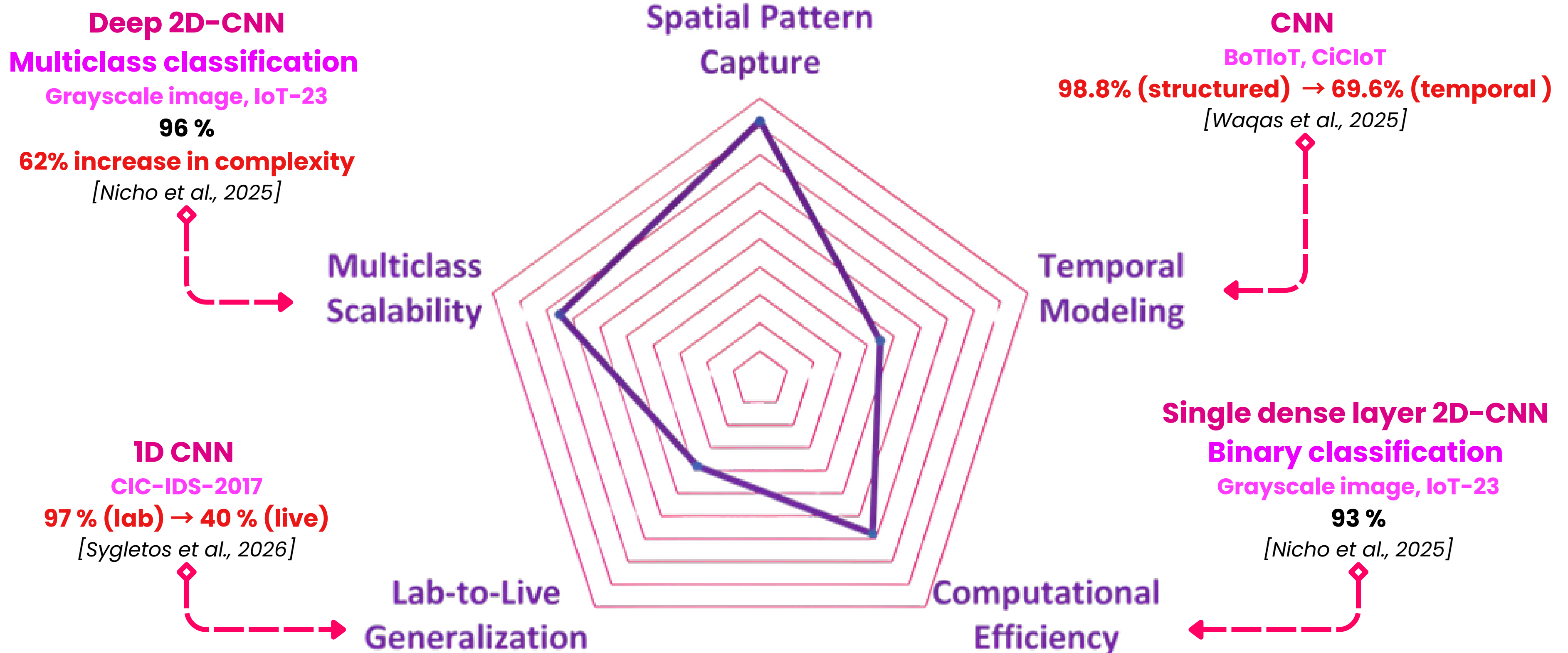
POOLING
FLATTENING



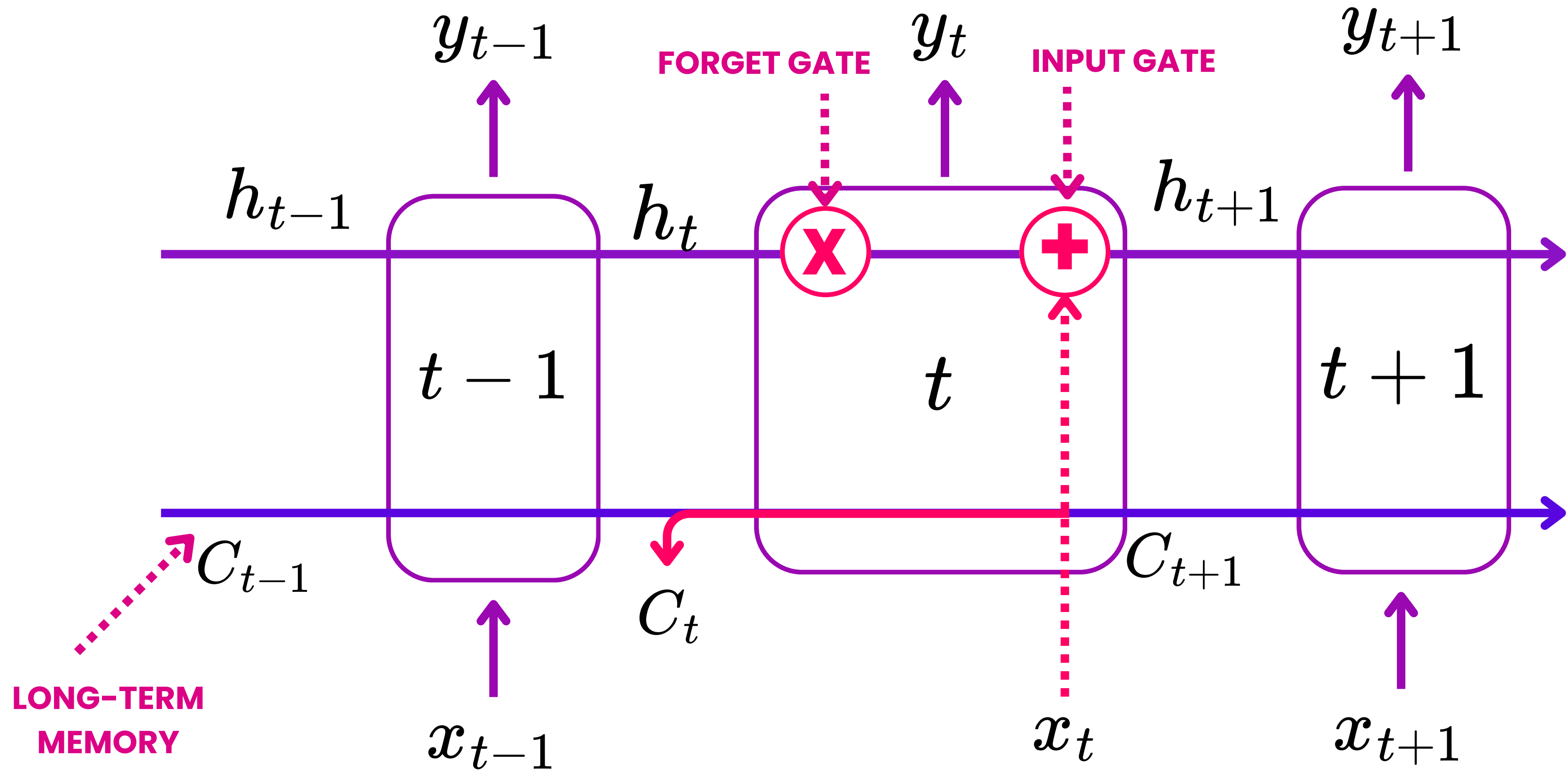
OUTPUT
FC



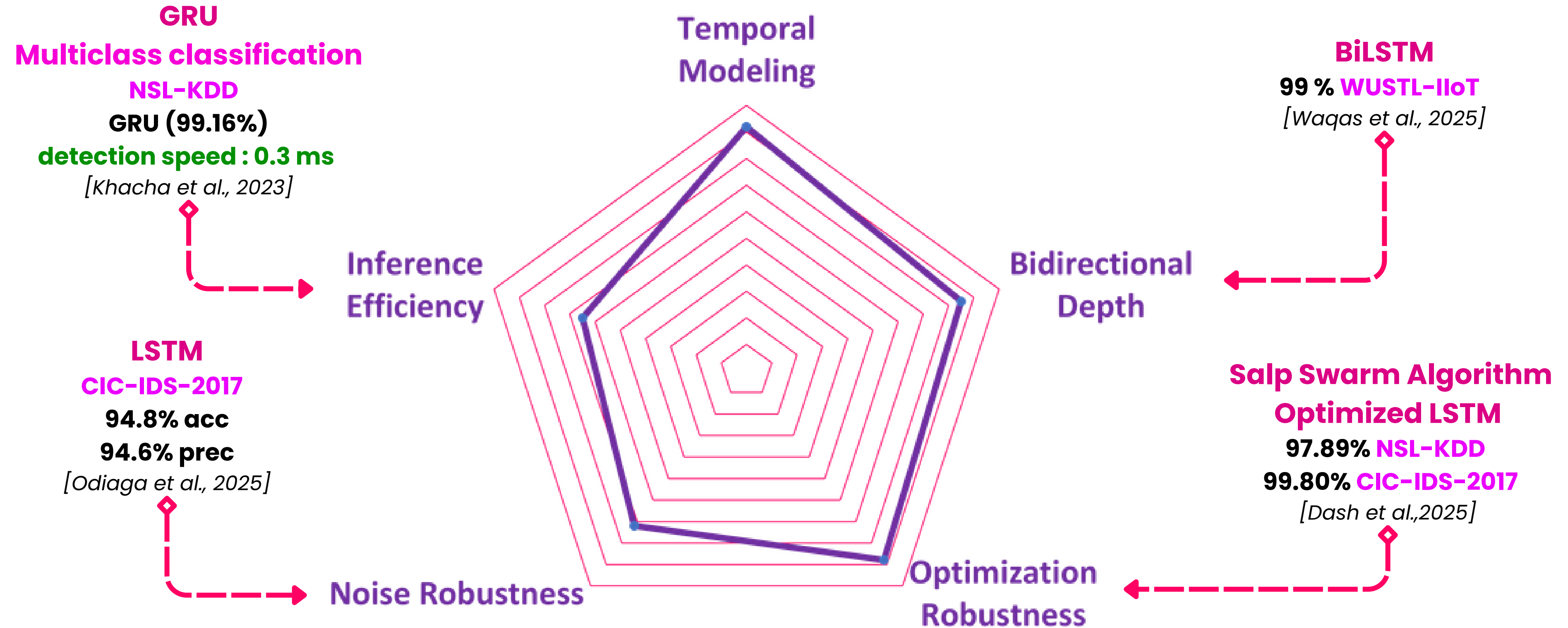
CNN LITERATURE REVIEW



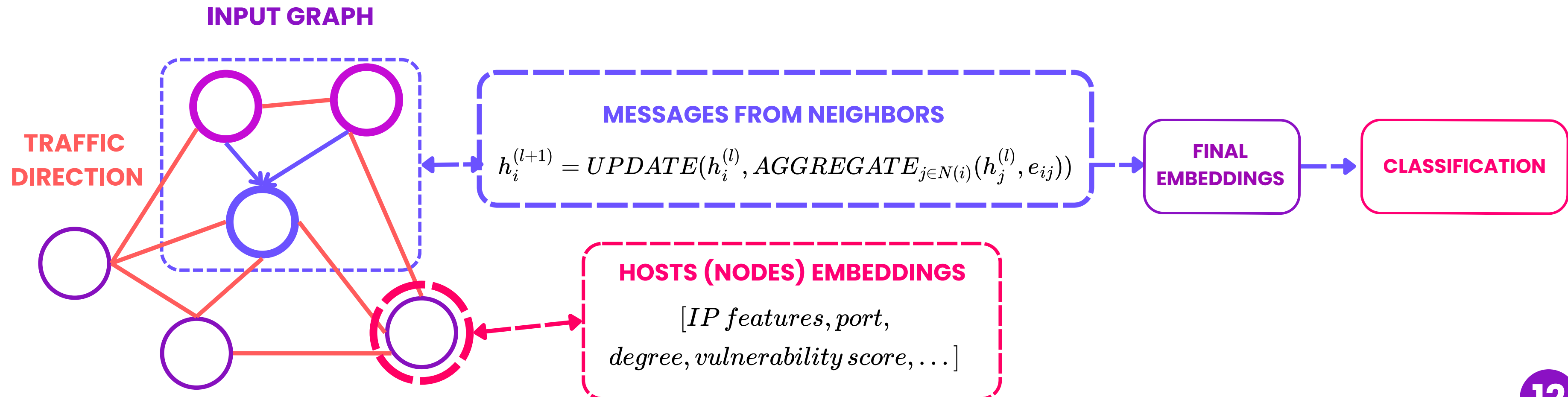
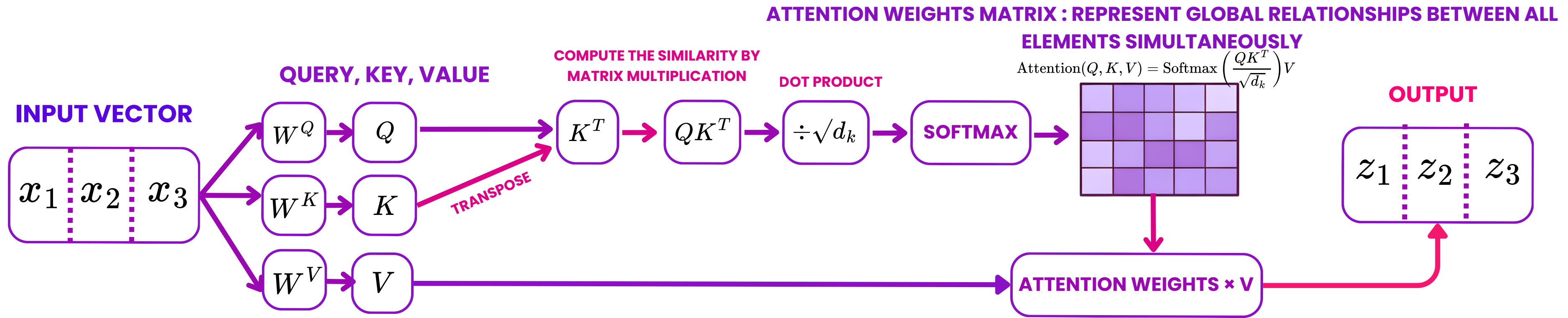
THE RECURRENT PARADIGM (LSTM)



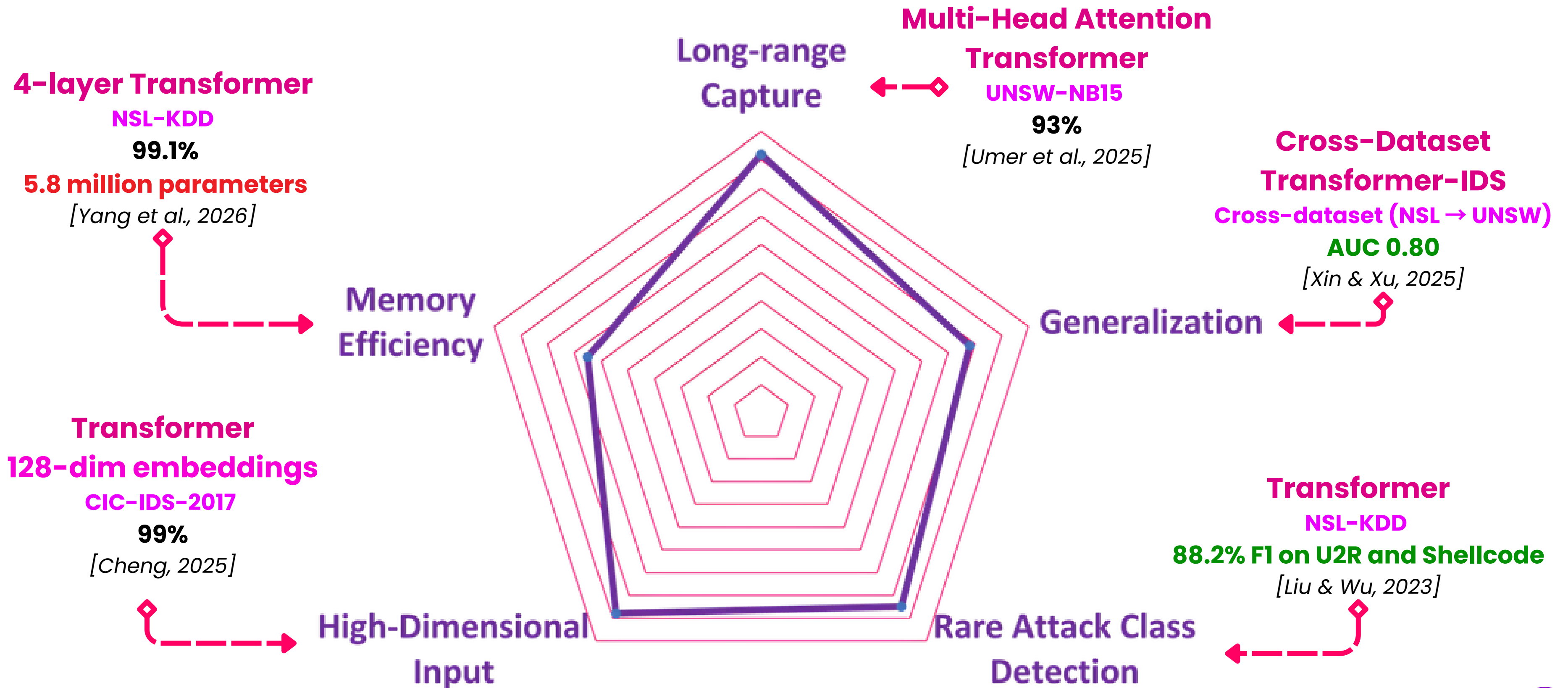
LSTM LITERATURE REVIEW



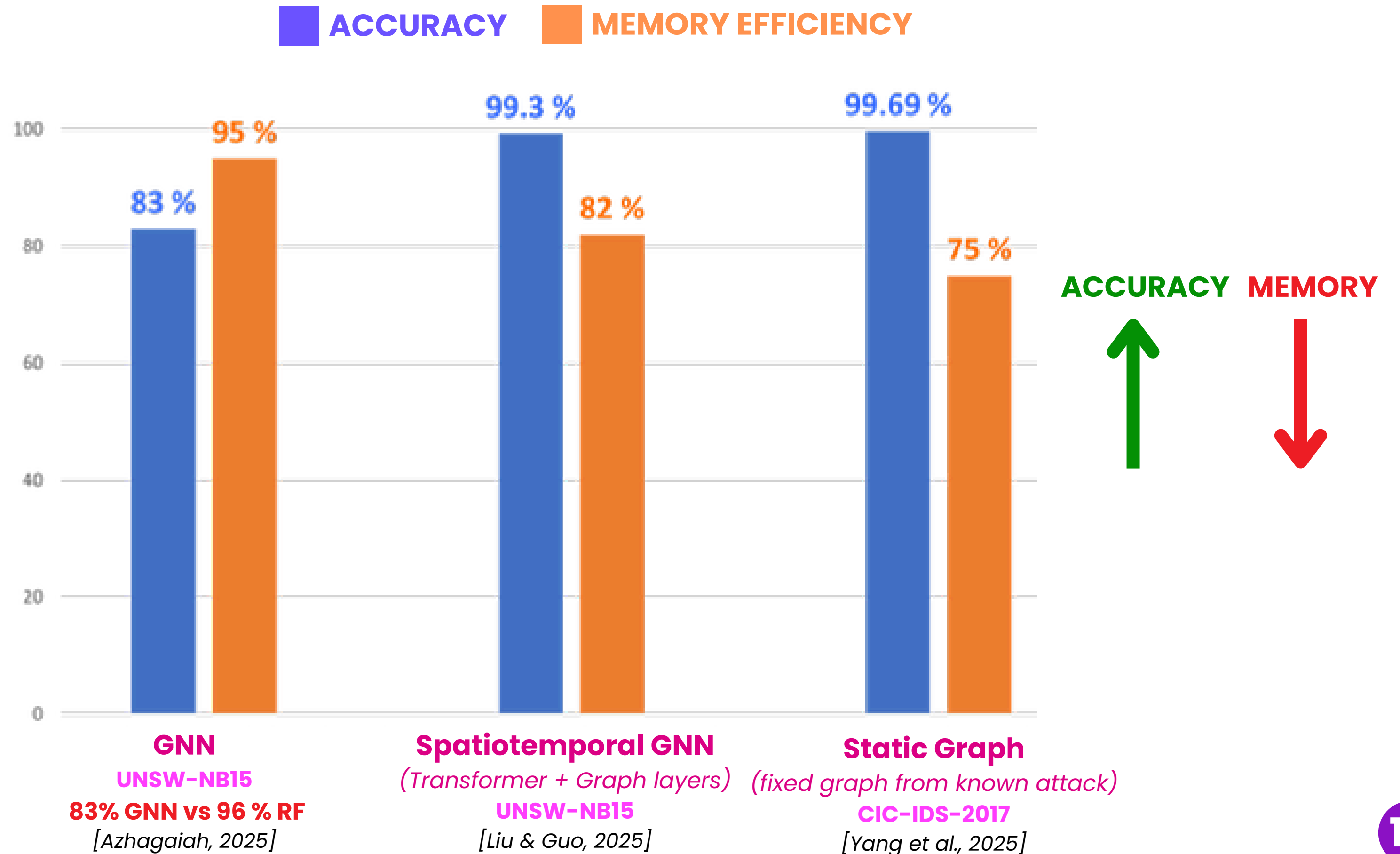
SELF-ATTENTION AND GRAPH-BASED ARCHITECTURES



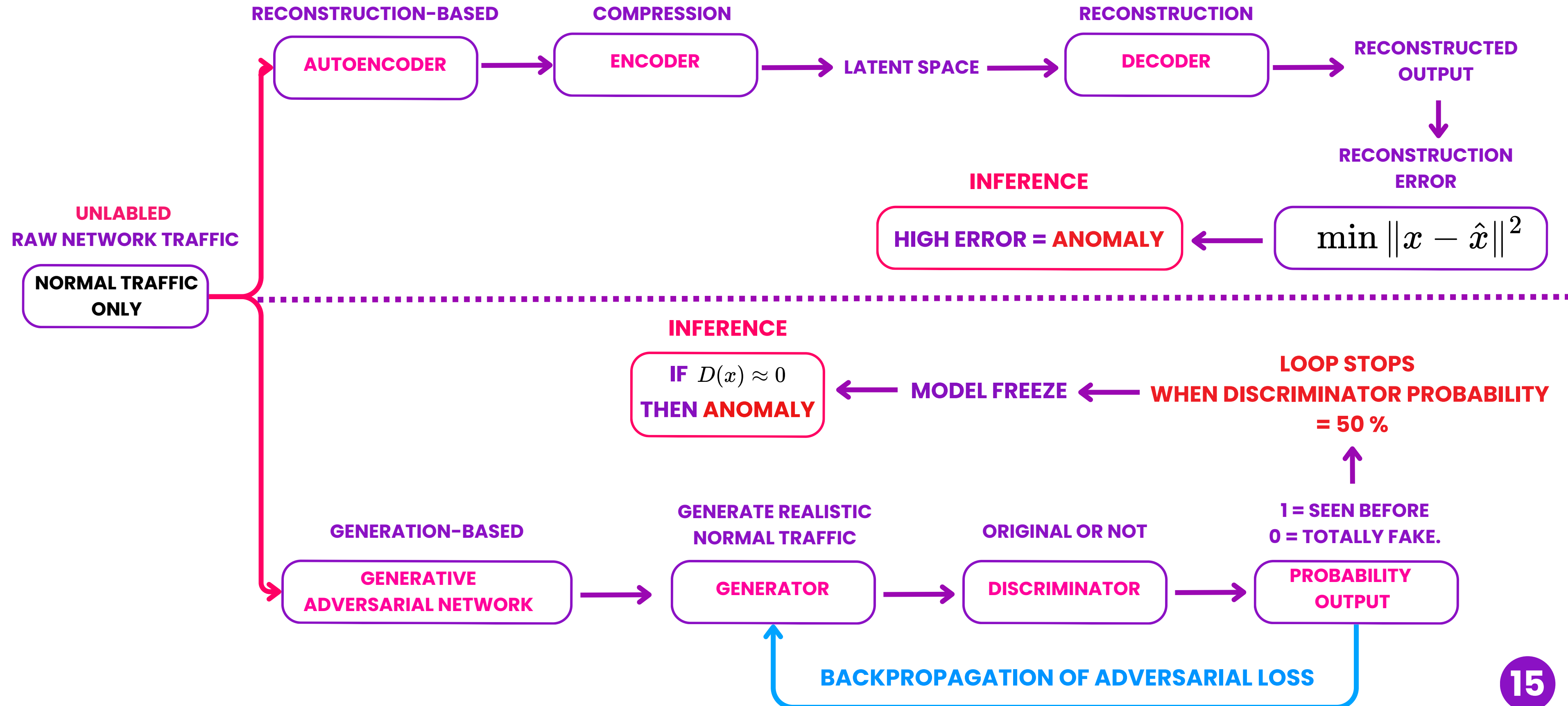
TRANSFORMER LITERATURE REVIEW



GNN LITERATURE REVIEW



UNSUPERVISED AND GENERATIVE MODELS



AUTOENCODERS REVIEW

Standard Autoencoder

Trained only on normal traffic

NSL-KDD

99.1%

[Hindy et al., 2020]

Zero-day

Sparse AE

Activate only critical neurons to
detect rare anomalies

UNSW-NB15

99.83%

[Pamena & Sugali, 2025]

Generalization

Standard AEs often
fail to distinguish
rare anomalies

[Pamena & Sugali,
2025]

Rare Anomaly
Detection

Masked Autoencoder

Self-supervised reconstruction

NetFlow

99.61% F1

[Singh & Jang-Jaccard, 2022]

Convolutional AE + LSTM

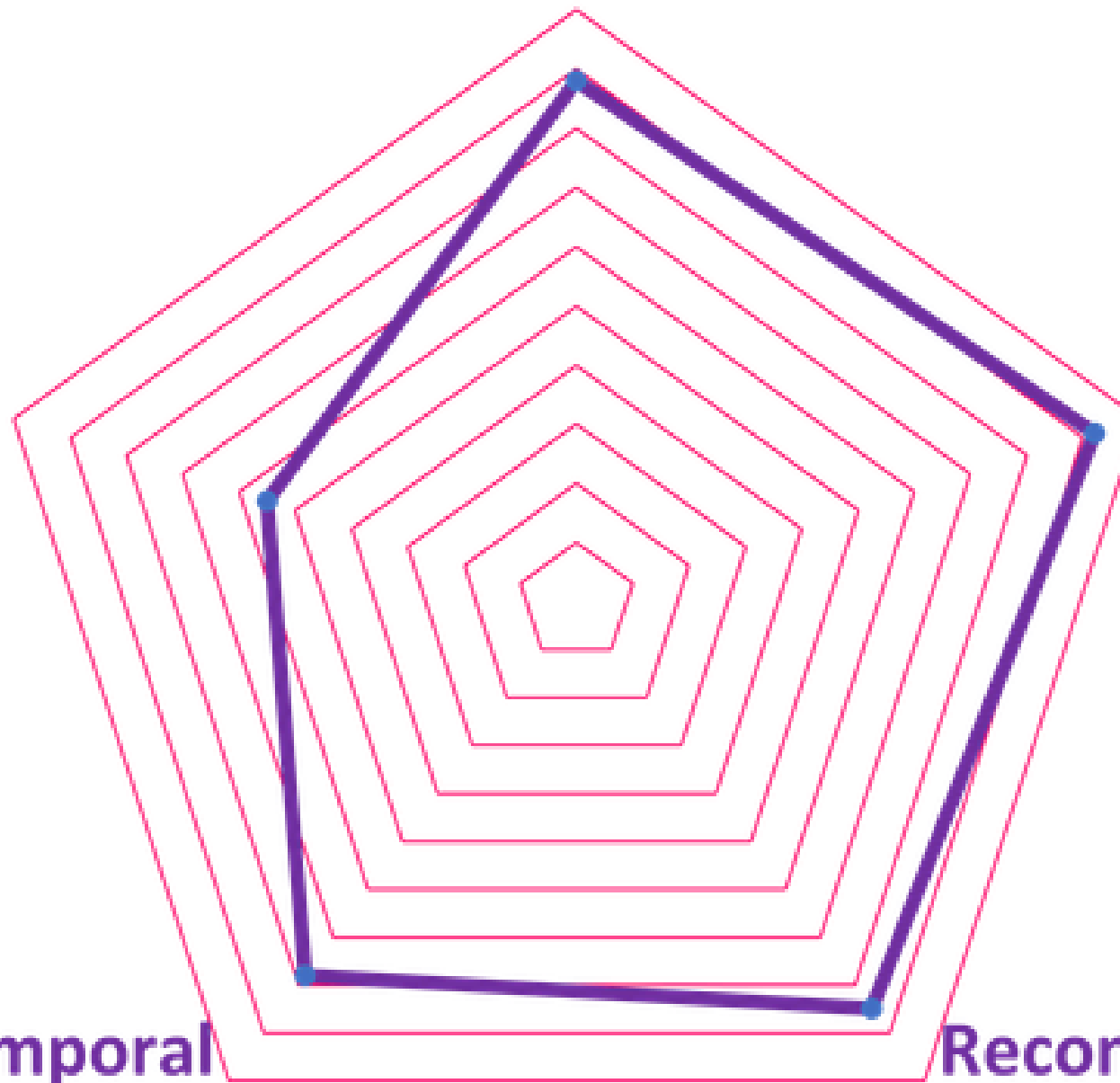
NSL-KDD

93.76%

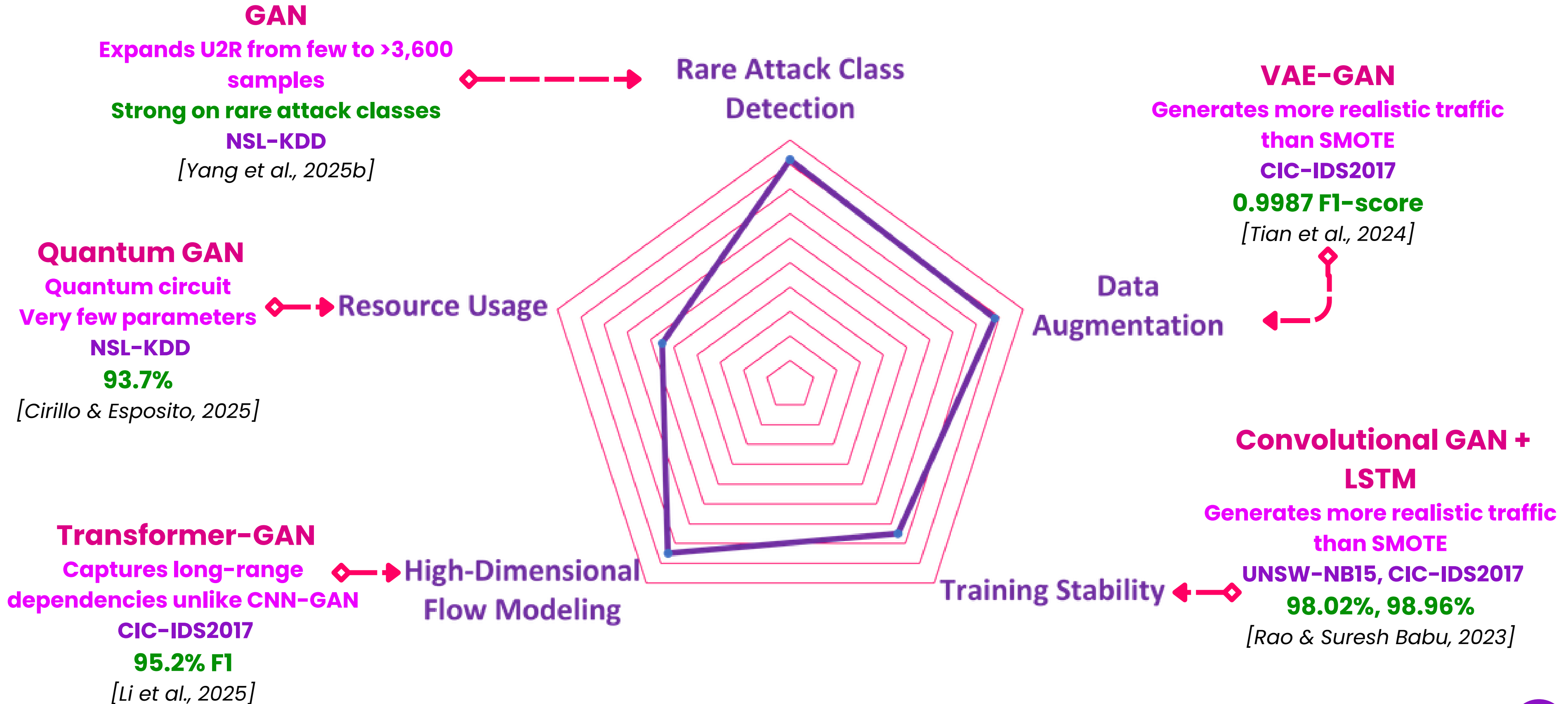
[Singh & Jang-Jaccard, 2022]

Spatiotemporal
Modeling

Reconstruction
Quality



GAN REVIEW



HYBRID ARCHITECTURES

SPATIOTEMPORAL HYBRIDS

MODEL	DATASET	PERFORMANCE
CAPSNET-BILSTM <i>[VIKRANTSHARMA, 2025]</i>	CIC-IDS2017	99%
CNN-LSTM <i>[FARABI ET AL., 2025]</i>	NSL-KDD	98% BINARY

RELATIONAL-TEMPORAL HYBRIDS

MODEL	DATASET	PERFORMANCE
GCN-BILSTM <i>[BIRADAR ET AL., 2025]</i>	UNSW-NB15 AS A GRAPH	97.5%
GNN-LSTM <i>[VITULYOVA ET AL., 2025]</i>	CICIDS2017	0.85 F1

ENSEMBLE HYBRIDS

MODEL	DATASET	PERFORMANCE
GAN + RESNET + CNN + LSTM <i>[YANG, 2024]</i>	NSL-KDD	99.6%
CNN-BILSTM-RF-LR <i>[ULLAH ET AL., 2025]</i>	CICIOT2023	99.2 %

COMPARATIVE SUMMARY OF REVIEWED IDS ARCHITECTURES

PAPER	DATASET	MODEL	RESULTS	CONTRIBUTION	LIMITATIONS
[ABDULLAH ET AL., 2025]	UNSW NB15	PCA+XGBOOST + SVM	98.0% ACC, 82.0% RECALL	HYBRID WITH PCA	F1-SCORE DROPS DRASTICALLY (0.40) ON HIGH-ENTROPY PACKETS
[WAQAS ET AL., 2025]	BOTIOT	1D CNN	98.8 % ACC	STRUCTURAL ROBUSTNESS ACROSS VARIABLE IOT ENV	LACKING SPATIAL STRUCTURE
[KHACHA ET AL., 2023]	EDGE- IIOTSET	GRU	98.46 % ACC	DEMONSTRATESGRU’S SUPERIORNOISEHANDLI NG	LATENCY
[XIN AND XU,2025]	NSL-KDD, UNSW NB15	CROSS-DATASET TRANSFORMER	0.80 AUC CROSS- DATASET	BRIDGED THE CROSS- DOMAIN GAP.	MASSIVE PARAMETER FOOTPRINT
[YANG ET AL., 2025A]	CIC-IDS 2017	GNN	99.69 % ACC, 99.83 % RECALL	INTEGRATES NETWORK TOPOLOGY WITH REAL- TIME TRAFFIC STATISTICS	SEVERE MEMORY ISSUES.

IDENTIFIED RESEARCH GAPS

1

BENCHMARK OVERFITTING VS. REAL-WORLD GENERALIZATION

2

ADVERSARIAL VULNERABILITY

3

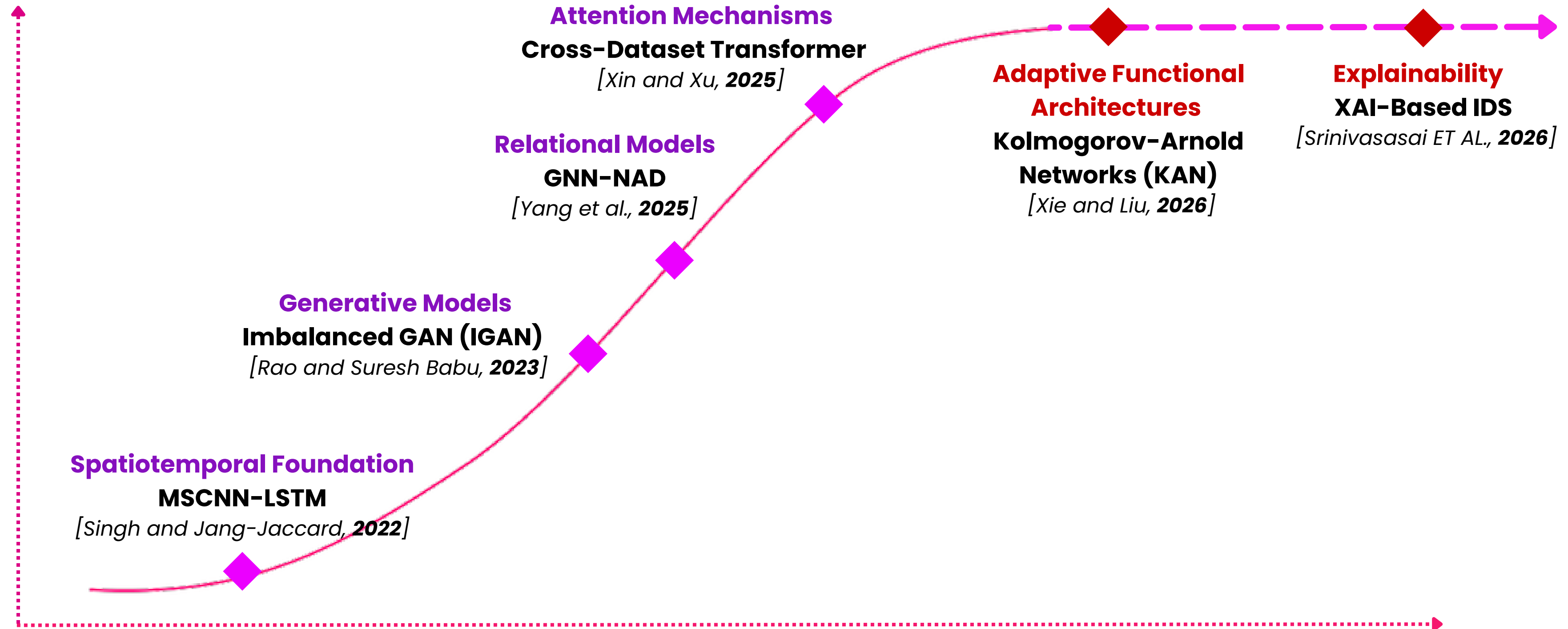
NO PUBLIC DATA ON INFERENCE LATENCY

4

NO COMPREHENSIVE FRAMEWORK FOR CONTINUOUS ADAPTATION

CONCLUSION : THE PATH FORWARD

ARCHITECTURAL MATURITY



TIMELINE (2021 - 2026)

THANK YOU FOR YOUR TIME AND ATTENTION.

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